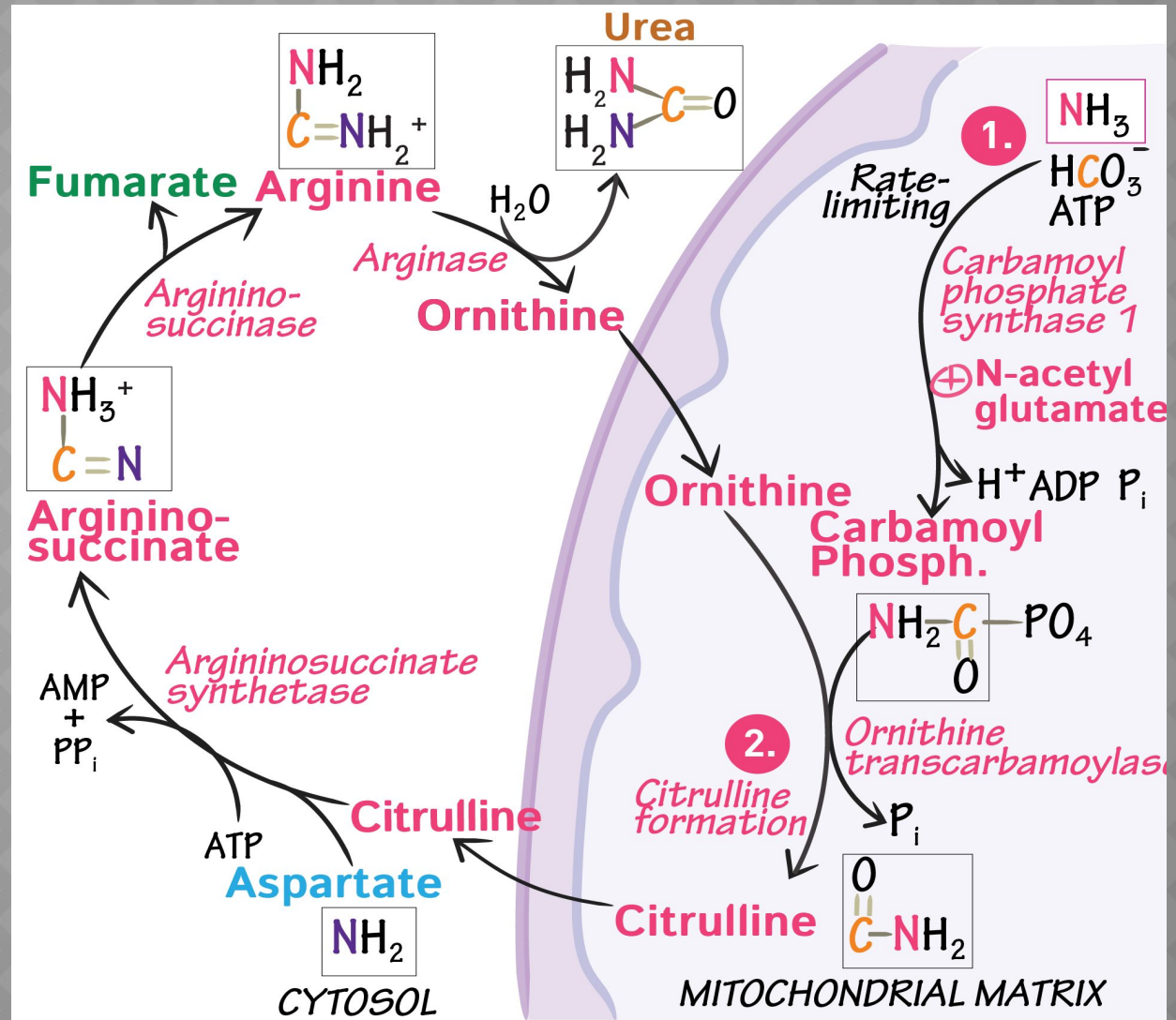


UREA CYCLE



OBJECTIVES

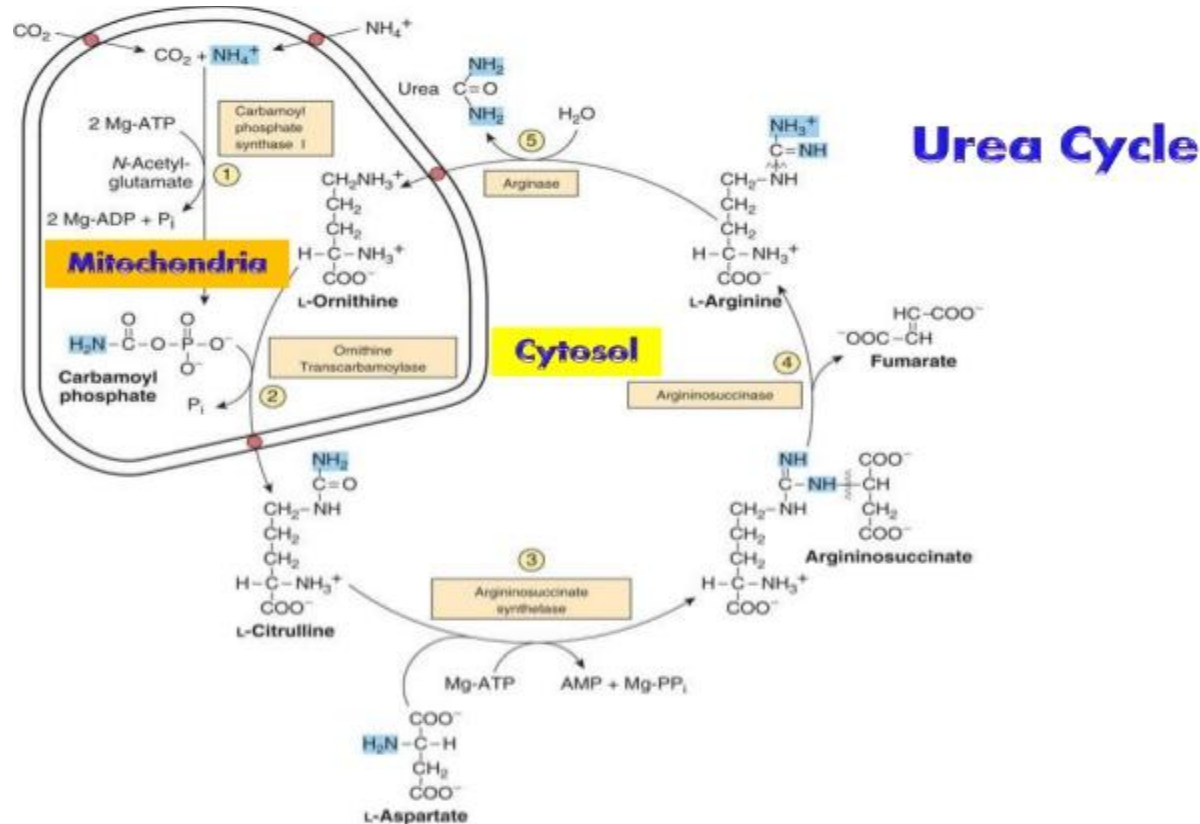
- ◉ Introduction
- ◉ Reactions of the cycle
- ◉ Fate of the urea
- ◉ Hyperammonemia

UREA CYCLE

- ◉ Urea is the major disposal form of amino groups derived from amino acids and accounts for about 90% of the nitrogen-containing components of urine.
- ◉ One nitrogen of the urea molecule is supplied by free ammonia and the other nitrogen by aspartate.
- ◉ Glutamate is the immediate precursor of both ammonia (through oxidative deamination by glutamate dehydrogenase) and aspartate nitrogen (through transamination of oxaloacetate by AST).
- ◉ The carbon and oxygen of urea are derived from CO_2 (as HCO_3^-).
- ◉ Urea is produced by the liver and then is transported in the blood to the kidneys for excretion in the urine.

REACTIONS OF THE CYCLE

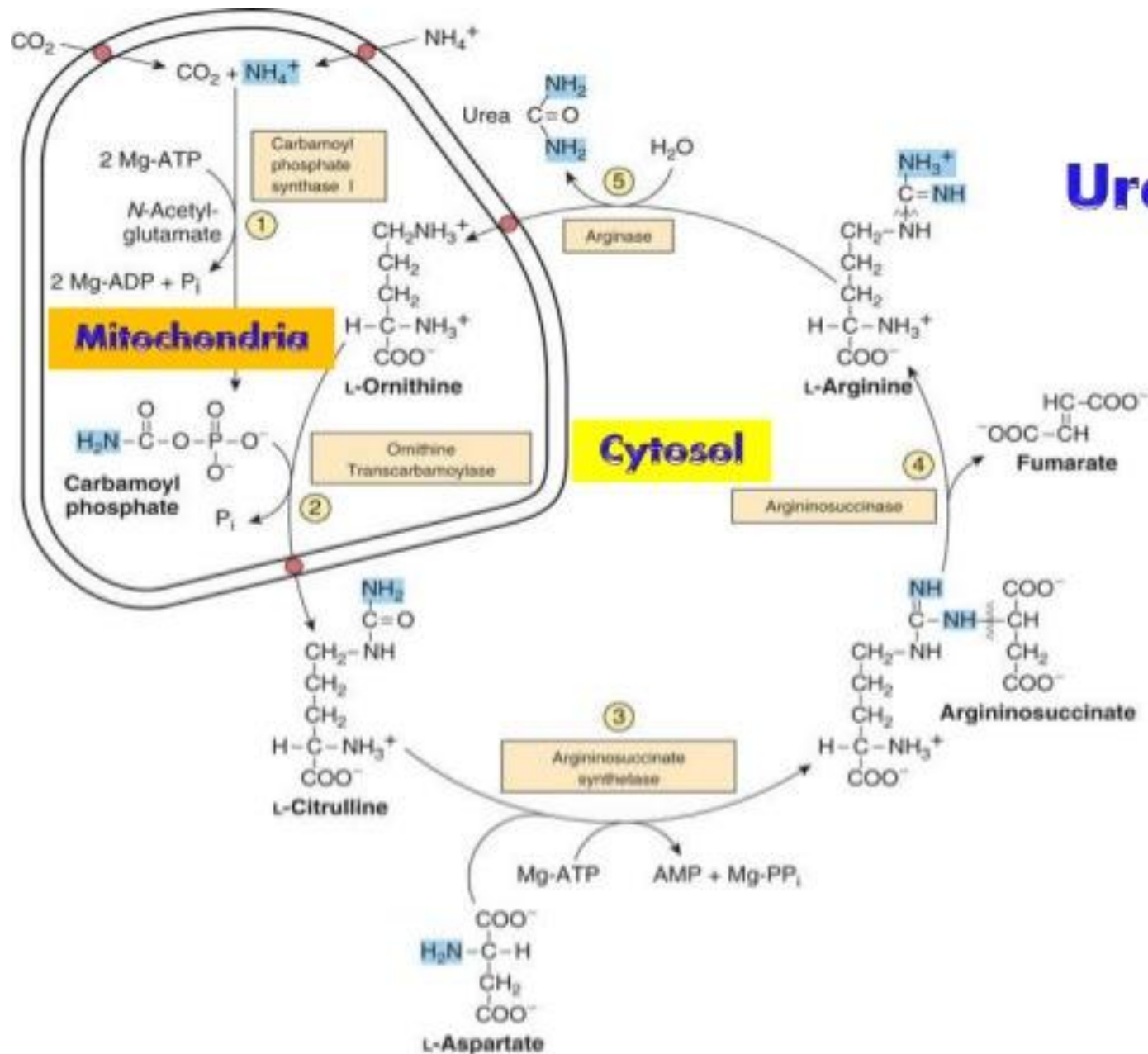
- The first two reactions leading to the synthesis of urea occur in the mitochondrial matrix, whereas the remaining cycle enzymes are located in the cytosol.



1. FORMATION OF CARBAMOYL PHOSPHATE:

- ◉ Formation of carbamoyl phosphate by carbamoyl phosphate synthetase I (CPS I) is driven by cleavage of two molecules of ATP.
- ◉ Ammonia incorporated into carbamoyl phosphate is provided primarily by the oxidative deamination of glutamate by mitochondrial glutamate dehydrogenase.
- ◉ Ultimately, the nitrogen atom derived from this ammonia becomes one of the nitrogens of urea.
- ◉ CPS I requires N-acetylglutamate as a positive allosteric activator

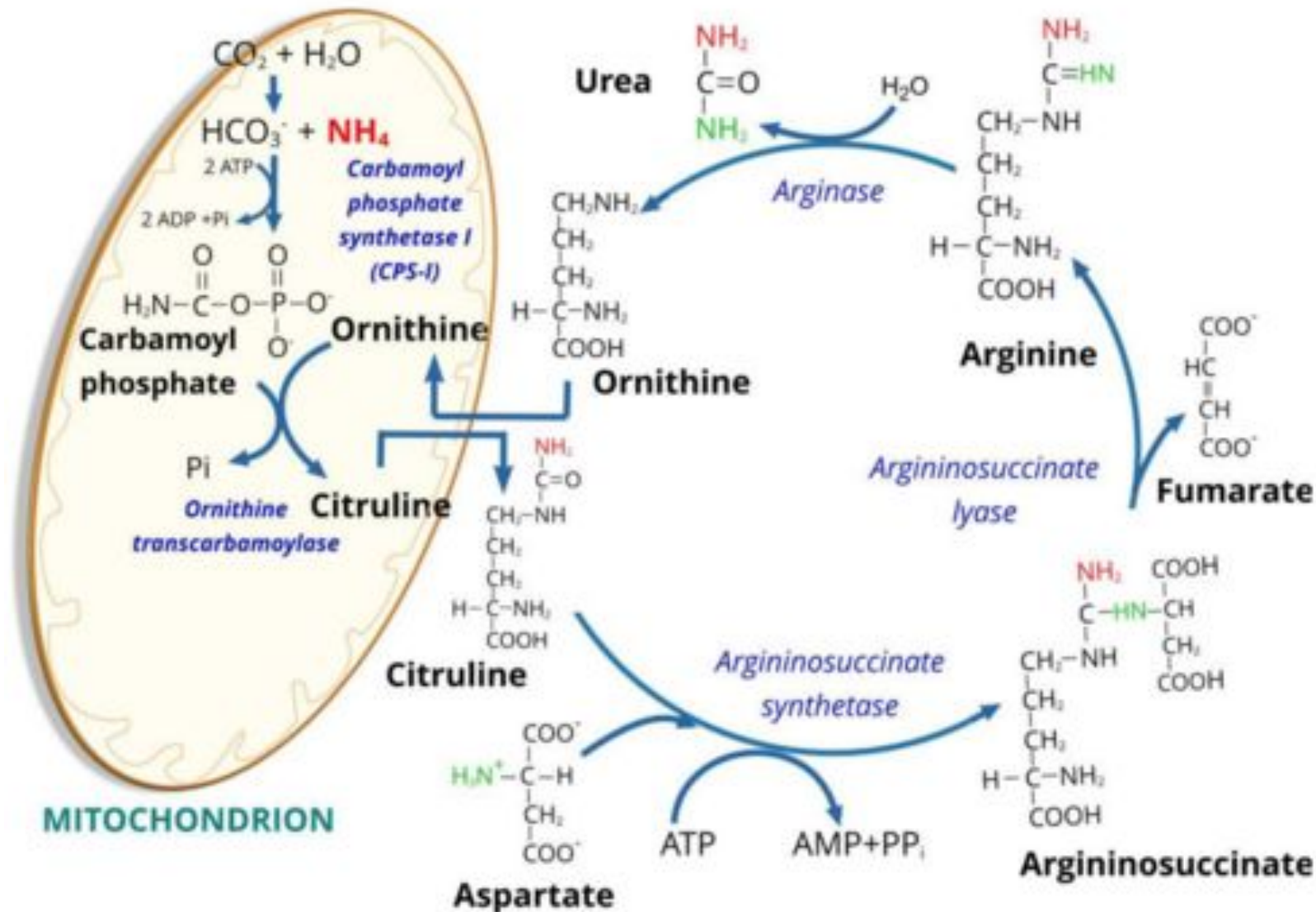
Urea Cycle



2. FORMATION OF CITRULLINE:

- ◉ The carbamoyl portion of carbamoyl phosphate is transferred to ornithine by ornithine transcarbamoylase (OTC) as the high-energy phosphate is released as inorganic phosphate.
- ◉ The reaction product citrulline is transported to the cytosol.
- ◉ Ornithine and citrulline are basic amino acids that participate in the urea cycle, moving across the inner mitochondrial membrane via a cotransporter.
- ◉ Ornithine is regenerated with each turn of the urea cycle, much in the same way that oxaloacetate is regenerated by the reactions of the citric acid cycle.

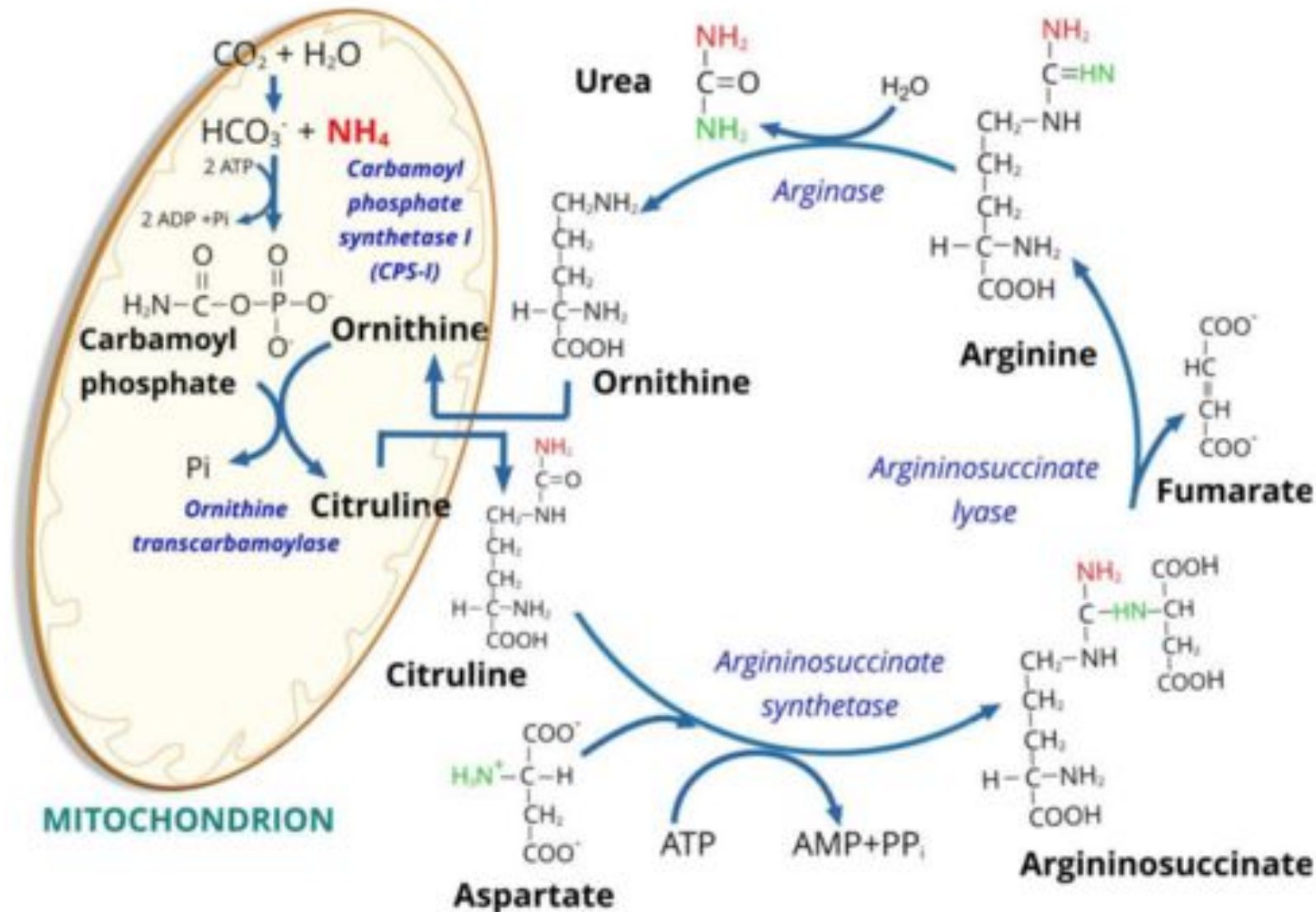
The Urea Cycle



3. SYNTHESIS OF ARGININOSUCCINATE:

- ◉ Argininosuccinate synthetase combines citrulline with aspartate to form argininosuccinate.
- ◉ The α -amino group of aspartate provides the second nitrogen that is ultimately incorporated into urea.
- ◉ The formation of argininosuccinate is driven by the cleavage of ATP to adenosine monophosphate and pyrophosphate.
- ◉ This is the third and final molecule of ATP consumed in the formation of urea.

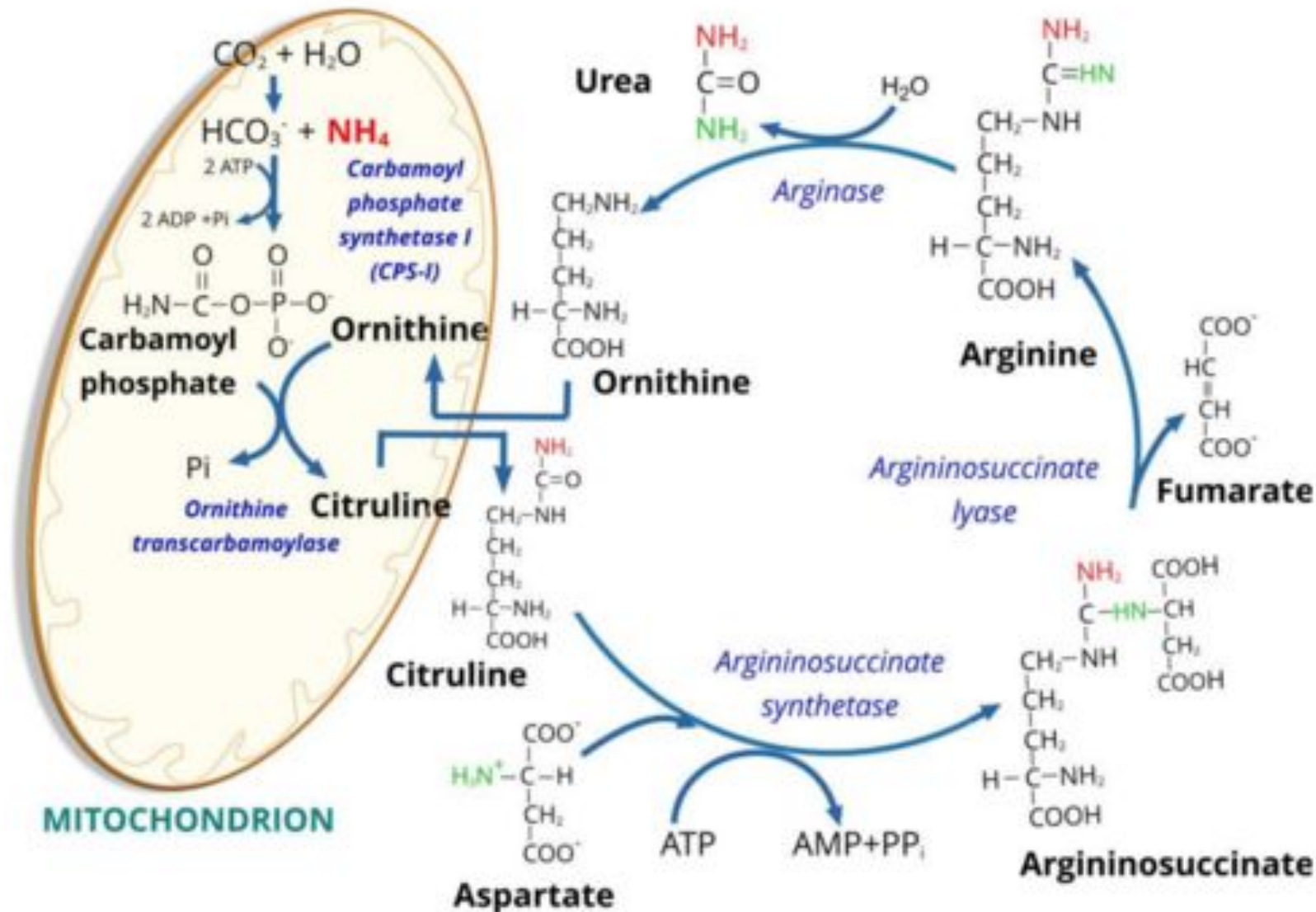
The Urea Cycle



4. CLEAVAGE OF ARGININOSUCCINATE:

- ◉ Argininosuccinate is cleaved by argininosuccinate lyase to yield arginine and fumarate.
- ◉ The arginine formed by this reaction serves as the immediate precursor of urea.
- ◉ Fumarate produced in the urea cycle is hydrated to malate, providing a link with several metabolic pathways.

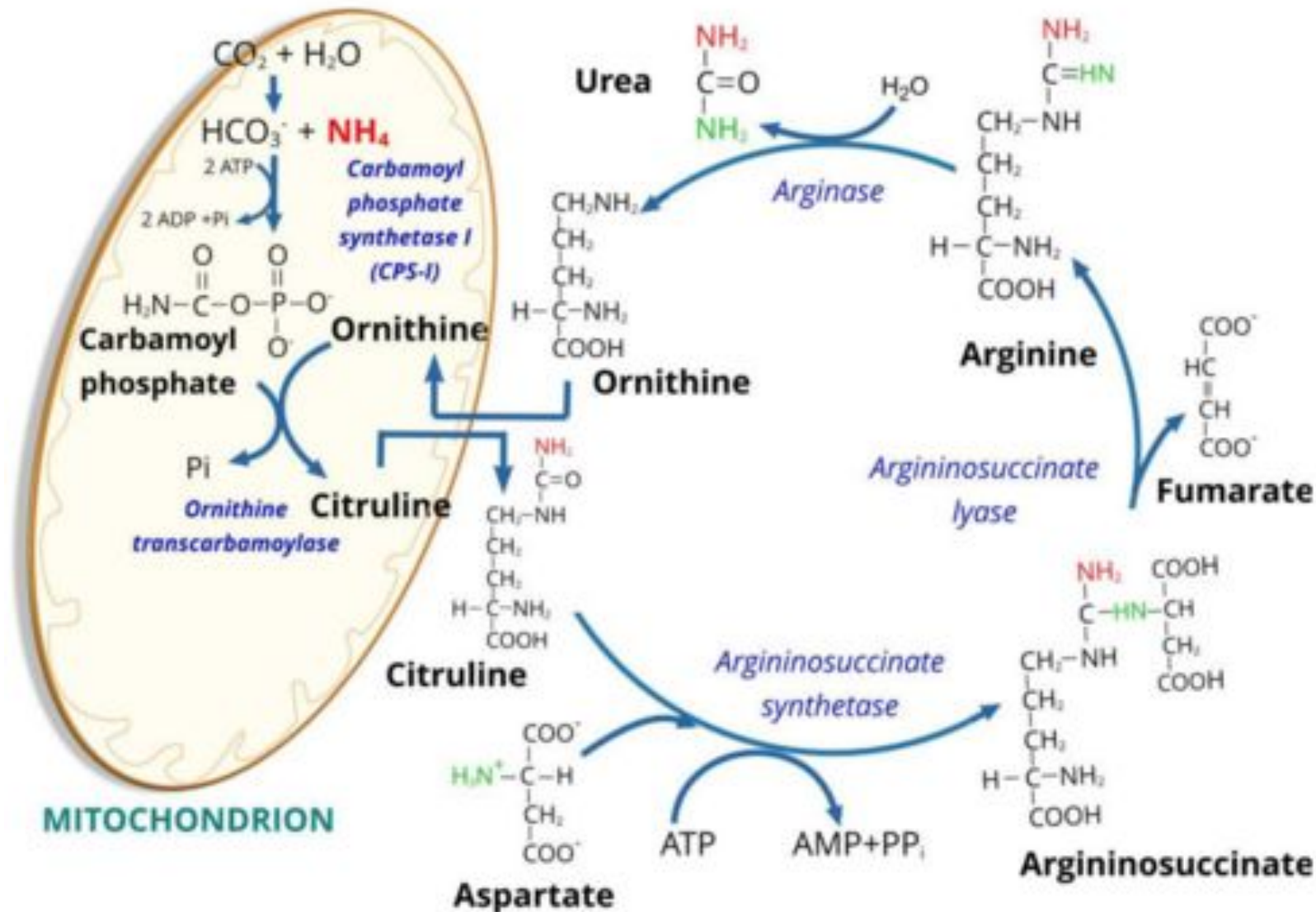
The Urea Cycle



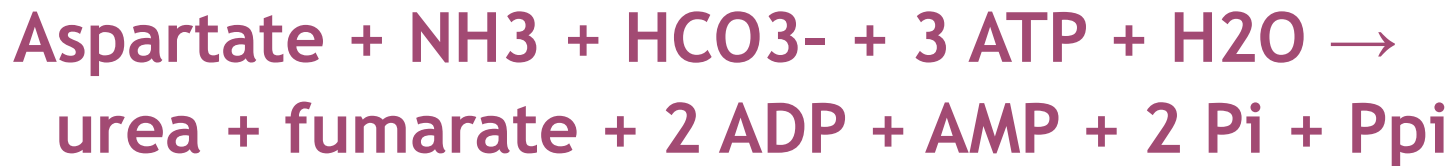
5. CLEAVAGE OF ARGININE TO ORNITHINE AND UREA:

- ◉ Arginase hydrolyzes arginine to ornithine and urea and is virtually exclusive to the liver.
- ◉ Therefore, only the liver can cleave arginine, thereby synthesizing urea.

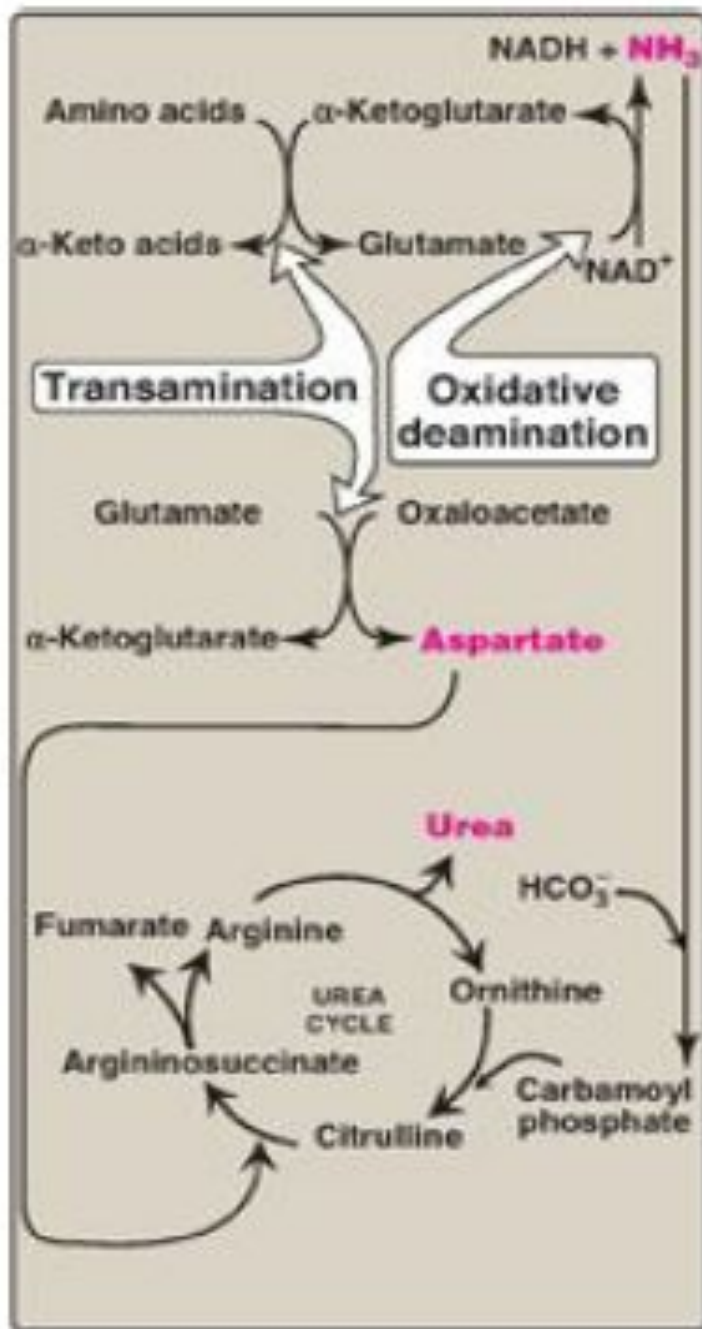
The Urea Cycle



OVERALL STOICHIOMETRY OF THE UREA CYCLE



- One nitrogen of the urea molecule is supplied by free NH_3 , and the other nitrogen by aspartate.
- Glutamate is the immediate precursor of both ammonia
- (through oxidative deamination by glutamate dehydrogenase) and aspartate nitrogen (through transamination of oxaloacetate by AST).
- In effect, both nitrogen atoms of urea arise from glutamate, which, in turn, gathers nitrogen from other amino acids.

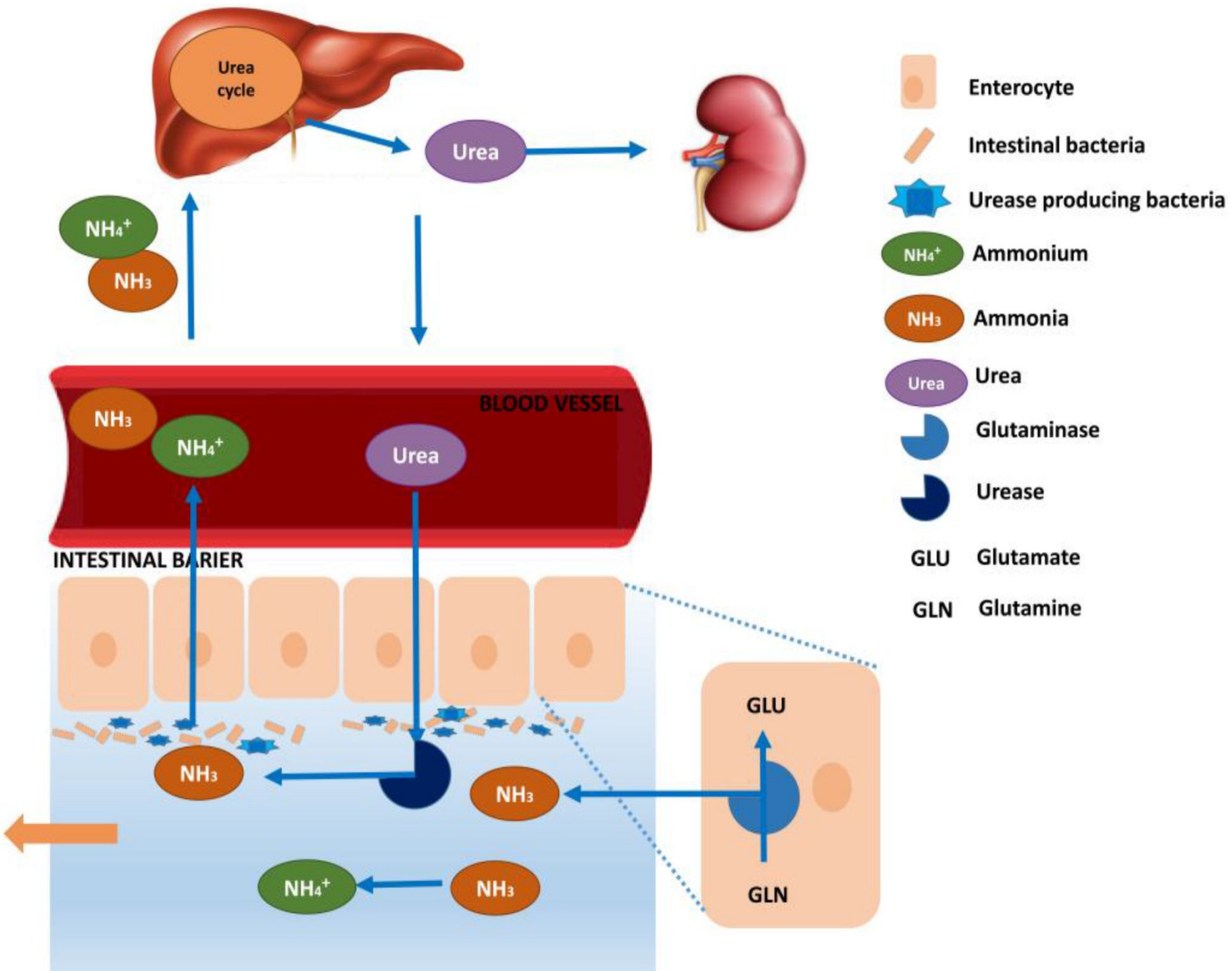


FATE OF UREA:

- ◎ Urea diffuses from the liver, and is transported in the blood to the kidneys, where it is filtered and excreted in the urine.
- ◎ A portion of the urea diffuses from the blood into the intestine and is cleaved to CO_2 and NH_3 by bacterial urease.
- ◎ This ammonia is partly lost in the feces and is partly reabsorbed into the blood.

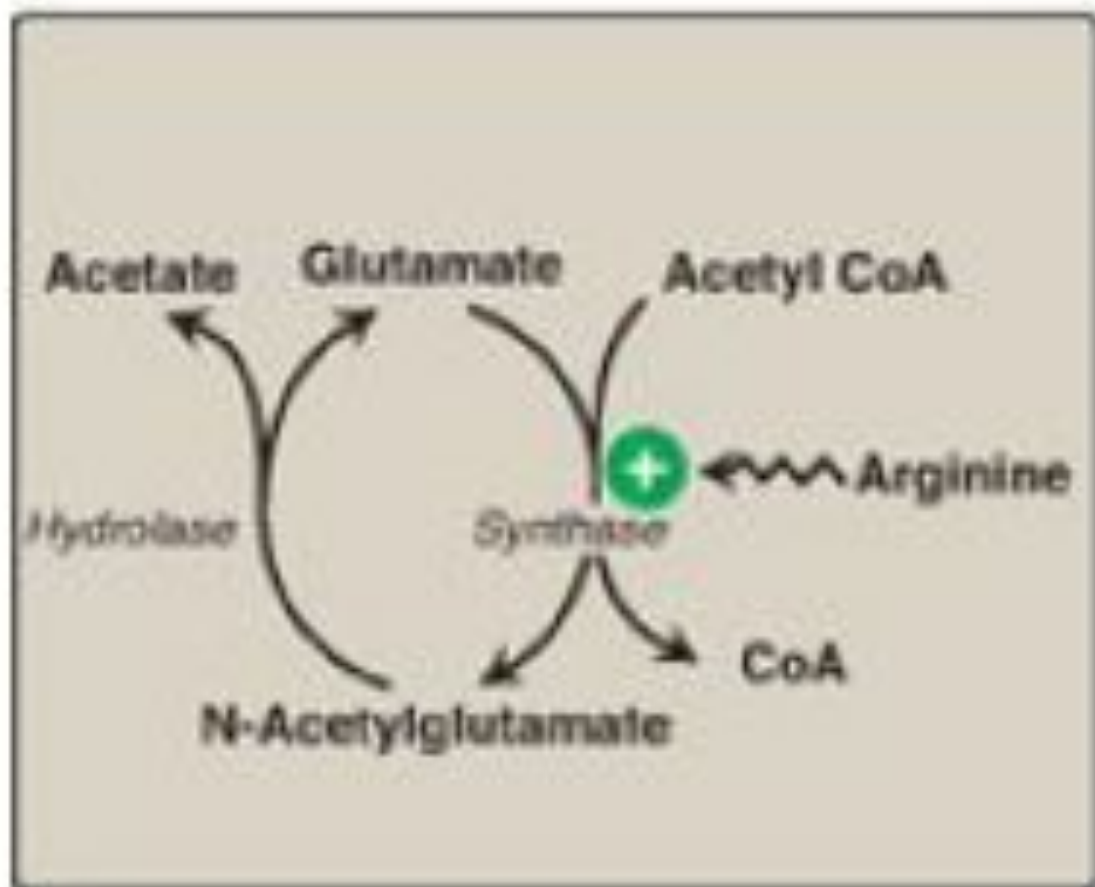
CONT..

- ◉ In patients with kidney failure, plasma urea levels are elevated, promoting a greater transfer of urea from blood into the gut.
- ◉ The intestinal action of urease on this urea becomes a clinically important source of ammonia, contributing to the hyperammonemia often seen in these patients.
- ◉ Oral administration of antibiotics reduces the number of intestinal bacteria responsible for this NH_3 production.



REGULATION OF THE UREA CYCLE

- ◉ N-Acetylglutamate (NAG) is an essential activator for CPS I, the rate-limiting step in the urea cycle.
- ◉ NAG is synthesized from acetyl CoA and glutamate by N-acetylglutamate synthase in a reaction for which arginine is an activator.
- ◉ The cycle is also regulated by substrate availability (short-term regulation) and enzyme induction (long term).



HYPERAMMONEMIA

- ◉ The capacity of the hepatic urea cycle exceeds the normal rates of ammonia generation, and the levels of serum ammonia are normally low.
- ◉ However, when liver function is compromised, due either to genetic defects of the urea cycle or liver disease, blood levels can rise.
- ◉ Such hyperammonemia is a medical emergency, because ammonia has a direct neurotoxic effect on the CNS.
- ◉ For example, elevated concentrations of ammonia in the blood cause the symptoms of ammonia intoxication, which include tremors, slurring of speech, somnolence (drowsiness), vomiting, cerebral edema, and blurring of vision.
- ◉ At high concentrations, ammonia can cause coma and death.

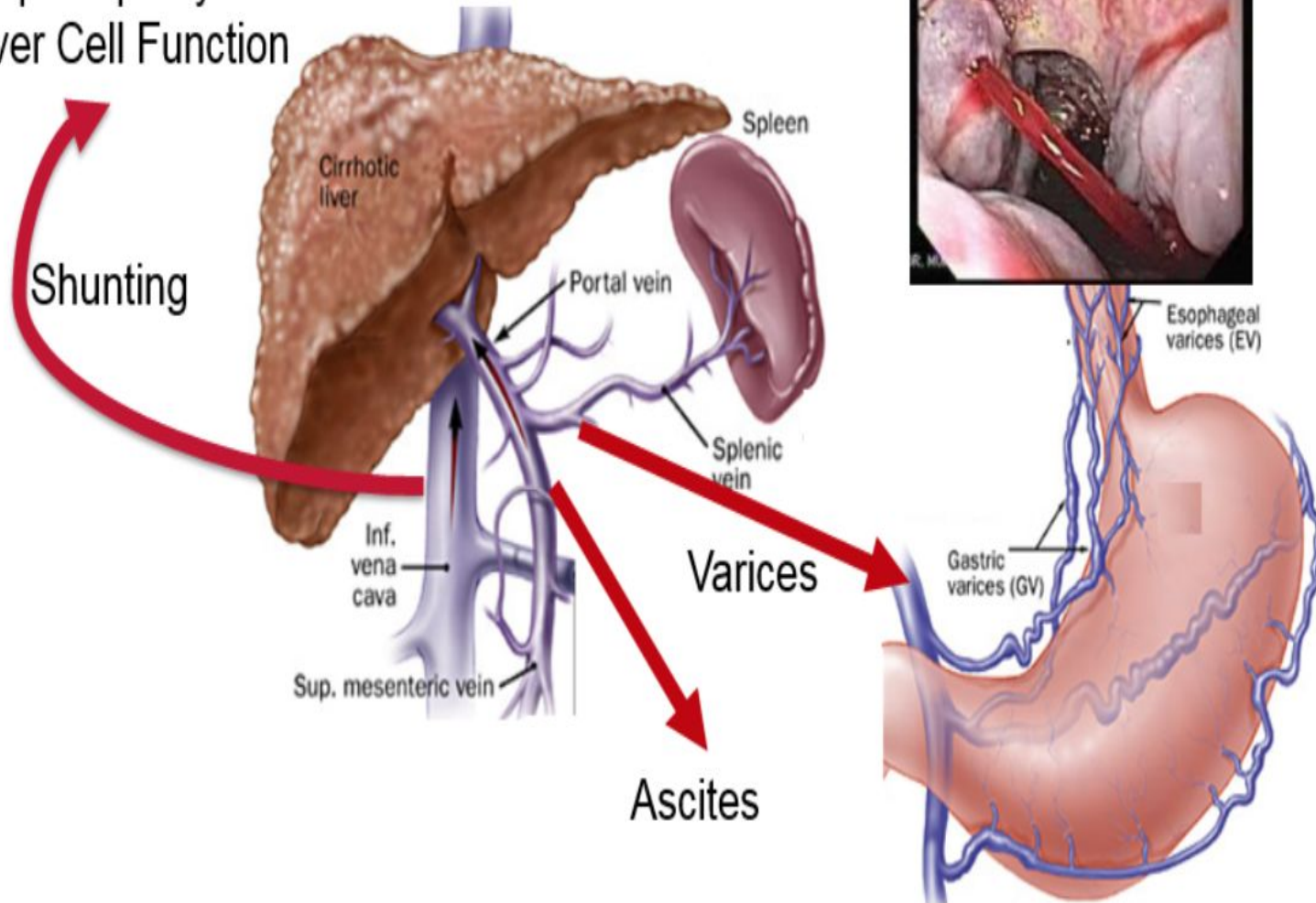
TYPES OF HYPERAMMONEMIA

- ◉ There are two major types of hyperammonemia

1. Acquired hyperammonemia:

- ◉ Liver disease is a common cause of hyperammonemia in adults and may be due, for example, to viral hepatitis or to hepatotoxins such as alcohol.
- ◉ Cirrhosis of the liver may result in formation of collateral circulation around the liver. As a result, blood is shunted directly into the systemic circulation and does not have access to the liver. Therefore, the conversion of ammonia to urea is severely impaired, leading to elevated levels of ammonia.

Encephalopathy
↓ Liver Cell Function



2. Congenital hyperammonemia:

- ◉ Genetic deficiencies of each of the five enzymes of the urea cycle have been described, with an overall incidence estimated to be 1:25,000 live births.
- ◉ X-linked ornithine transcarbamylase deficiency is the most common of these disorders, predominantly affecting males, although female carriers may become symptomatic.
- ◉ All of the other urea cycle disorders follow an autosomal recessive inheritance pattern.

CONT..

- In each case, the failure to synthesize urea leads to hyperammonemia during the first weeks following birth.
- Historically, urea cycle defects had high morbidity (neurologic manifestations) and mortality.

THANK YOU